

Efficiency vs. Accuracy: Finding the Breakeven Point Between Fine-Tuned SLMs and Few-Shot LLMs

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Abstract

Running a large language model for every inference request is expensive, but smaller models need labeled training data before they can compete. This paper asks a direct question: how many labeled examples does it take before fine-tuning a small model beats few-shot prompting a large one? We call this threshold the *Breakeven Point* and measure it across four NLP tasks: GoEmotions (12-class emotion recognition), AG News (4-class topic classification), SNLI (3-class natural language inference), and CoLA (binary linguistic acceptability). We fine-tune DistilBERT on training sets ranging from 50 to 1200 examples and compare against Mistral-7B-Instruct evaluated under six different few-shot configurations. The answer turns out to depend heavily on the task. DistilBERT crosses the Mistral baseline at around 450 examples on GoEmotions, still has not caught up on AG News or SNLI within our tested range, and plateaus well below Mistral on CoLA even at 1000 examples. The results give practitioners a concrete data threshold to work with when deciding whether annotation investment is worth it.

1 Introduction

Suppose a company wants to classify customer support tickets by category and expects a million requests per day. They can prompt a large language model with a handful of examples and get decent results immediately, but the per-query API cost scales linearly with volume. Alternatively, they can collect a few hundred labeled examples, fine-tune a lightweight model, and run inference cheaply forever. The question is: at what point does the second option start making sense?

This is not just a cost question. It is also a performance question. Large instruction-tuned models like Mistral-7B (Jiang et al., 2023) are strong few-shot learners because their pre-training has exposed them to enormous amounts of text covering nearly

every topic and linguistic structure. Smaller distilled models like DistilBERT (Sanh et al., 2019) are far cheaper to run but start from a much weaker position on any specific task until they are fine-tuned.

We formalize this as the **Breakeven Point**: the minimum number of labeled training examples n^* at which a fine-tuned DistilBERT model matches or exceeds the Macro F1 score of few-shot Mistral-7B-Instruct. To see how much this threshold shifts across different types of problems, we run experiments on four tasks that vary in label count, linguistic difficulty, and how much prior knowledge an LLM can bring to bear: GoEmotions (Demszky et al., 2020), AG News (Zhang et al., 2015), SNLI (Bowman et al., 2015), and CoLA (Warstadt et al., 2019).

We also run a sensitivity analysis on how many few-shot examples Mistral actually needs to reach its ceiling, and test whether the emotion results hold up when we swap in different subsets of emotion categories. The goal throughout is to give practical guidance for the annotation-vs-prompting decision, grounded in actual measured numbers rather than general intuition.

2 Related Work

Brown et al. (2020) showed that scaling language models produces strong few-shot performance across many tasks, establishing in-context learning as a practical alternative to fine-tuning. Sanh et al. (2019) showed that a distilled version of BERT retains 97% of language understanding capacity at 60% of the size and speed, making it a natural candidate for low-cost deployment. Lu et al. (2022) documented that few-shot performance is sensitive to which specific examples are chosen for the prompt, which motivated us to average across multiple random demonstration subsamples rather than relying on a single trial.

Prior work has studied data efficiency for fine-tuning in isolation, and separately studied the properties of in-context learning, but direct comparisons between the two under a controlled budget grid across multiple task types are less common. The closest work we are aware of is [?](#), who compared few-shot and fine-tuned models but focused on a single task type. Our work maps the full data efficiency curve across four benchmarks with different structural properties, which lets us identify what determines where the crossover happens.

3 Methodology

3.1 Task and Dataset Selection

We chose four tasks specifically to cover a range of conditions that might affect where the breakeven falls. GoEmotions has 12 output classes and requires distinguishing between subtle, overlapping emotional states in informal Reddit text. AG News has only 4 classes and involves clear-cut topical distinctions that are well-represented in any large pre-training corpus. SNLI requires compositional reasoning over pairs of sentences rather than classifying a single input. CoLA is binary but demands sensitivity to syntactic phenomena that are hard to articulate, even for large models. Together these four tasks let us examine how label count, task structure, and LLM prior knowledge all interact with the breakeven threshold.

3.2 Models

Fine-Tuning Model. We use DistilBERT-base-uncased ([Sanh et al., 2019](#)), a 66M-parameter distilled transformer. For each task and training size, we fine-tune a classification head with a learning rate of 2×10^{-5} , batch size of 16, and up to 5 training epochs with early stopping (patience = 1) based on validation Macro F1.

Few-Shot Model. We use Mistral-7B-Instruct-v0.2 ([Jiang et al., 2023](#)), a 7B-parameter instruction-tuned model. For each evaluation, we construct a prompt with k stratified demonstrations sampled from the training set, followed by the test example. The model generates a label token via greedy decoding. No weight updates occur.

3.3 Evaluation Protocol

All experiments use Macro F1 as the primary metric to handle class imbalance. For fine-tuning, we run $N = 4$ trials per configuration (or $N = 2$ for extended sweeps) using different random seeds for

both data subsampling and model initialization. For few-shot evaluation, we run the same number of trials with different demonstration subsamples per trial, following [Lu et al. \(2022\)](#) to reduce sensitivity to prompt selection. We report mean and standard deviation across trials.

We use fast evaluation subsets of 90–120 stratified examples for the test and validation sets across all tasks. This was a practical necessity given our compute constraints running experiments on Google Colab with limited GPU time. The tradeoff is that our F1 estimates carry more sampling noise than a full-dataset evaluation would. We discuss this further in the limitations section.

3.4 Data Budget Grid

For primary experiments, we evaluate DistilBERT at $n \in \{50, 150, 450, 600\}$ training examples with Mistral at $k = 36$ demonstrations. For extended sensitivity analysis, we expand to $n \in \{50, 150, 300, 450, 600, 750, 1000, 1200\}$ and $k \in \{12, 24, 36, 48, 60, 96\}$.

4 Experiments and Results

4.1 GoEmotions: Emotion Recognition

Setup. We filter GoEmotions ([Demszky et al., 2020](#)) to 12 single-label emotion classes: admiration, amusement, anger, annoyance, approval, curiosity, disappointment, disapproval, gratitude, joy, love, and sadness. We use 120 stratified test and validation examples.

Primary Results. Table 1 and Figure 1 show the primary results. Mistral achieves a mean F1 of 0.370 with $k = 36$ demonstrations. DistilBERT crosses this at $n = 450$ (F1 = 0.389), which is roughly 37.5 labeled examples per class. At $n = 600$, DistilBERT reaches F1 = 0.511, which is 38% above the Mistral baseline.

Model	Config	Macro F1
Mistral (few-shot)	$k = 36$	0.370 ± 0.027
DistilBERT	$n = 50$	0.041 ± 0.021
DistilBERT	$n = 150$	0.129 ± 0.060
DistilBERT	$n = 450$	0.389 ± 0.040
DistilBERT	$n = 600$	0.511 ± 0.034

Table 1: GoEmotions primary results. Breakeven at $n \approx 450$.

Extended Sensitivity Analysis. Figure 2 shows how the breakeven shifts as we vary k . Mistral’s

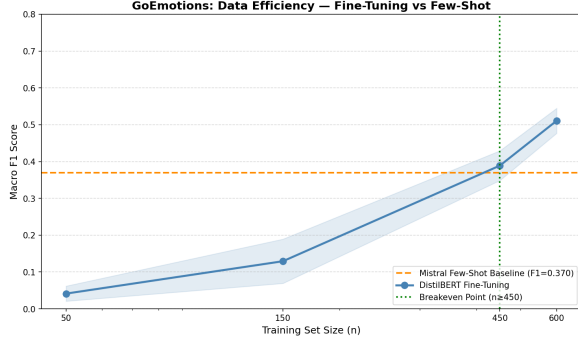


Figure 1: GoEmotions Data Efficiency Curve. DistilBERT crosses the Mistral baseline at $n = 450$, with confidence bands showing ± 1 standard deviation.

performance climbs from $F1 = 0.260$ at $k = 12$ to $F1 = 0.422$ at $k = 48$, then flattens out: $k = 60$ and $k = 96$ both score slightly lower (0.419 and 0.408), suggesting the model has hit its ceiling on this task regardless of how many prompt examples it sees. DistilBERT’s curve keeps rising across the full range, reaching $F1 = 0.644$ at $n = 1200$. The practical implication is that adding more prompt examples beyond $k = 48$ does not meaningfully raise the bar DistilBERT needs to clear, so the breakeven point is more stable than it might initially appear.

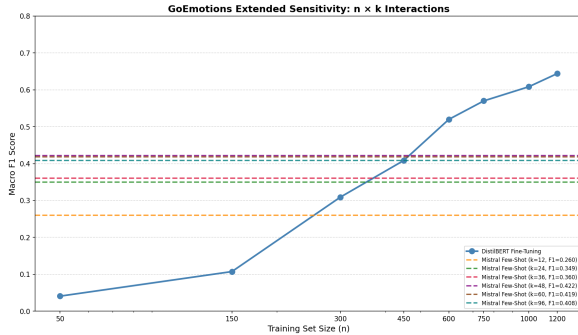


Figure 2: Extended sensitivity analysis on GoEmotions showing the $n \times k$ interaction. Multiple Mistral baselines (varying k) are plotted against the DistilBERT fine-tuning curve.

Emotion Subset Generalization. We tested whether the breakeven pattern holds when we swap in different emotion categories. We constructed two 12-class subsets: a Positive subset (admiration, amusement, approval, caring, desire, excitement, gratitude, joy, love, optimism, pride, relief) and a Negative subset (anger, annoyance, disappointment, disapproval, disgust, embarrassment, fear, grief, nervousness, remorse, sadness, confusion). Figure 3 shows the comparison. Mistral scores slightly higher on the Negative subset ($F1 = 0.431$

vs. 0.401 for Positive). In both cases, DistilBERT has not reached breakeven at $n = 450$: it achieves $F1 = 0.347$ on Positive and $F1 = 0.273$ on Negative. The gap is larger on Negative, which makes sense given the semantic overlap between labels like anger vs. annoyance and disappointment vs. disapproval. These fine distinctions require more data to learn than a model trained on only 37 examples per class can reliably make.

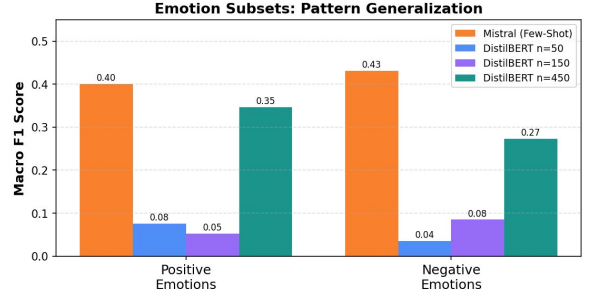


Figure 3: Emotion subset comparison: Positive vs. Negative 12-class subsets. Mistral few-shot performance and DistilBERT at $n \in \{50, 150, 450\}$ are shown for each subset.

4.2 AG News: Topic Classification

Setup. AG News (Zhang et al., 2015) contains short news articles across four topics: World, Sports, Business, and Science/Technology. We use 120 stratified test and validation examples, fine-tuning sizes $n \in \{20, 100, 400\}$, and $k = 12$ few-shot demonstrations, adjusted downward because 4-class prompts are shorter.

Results. Table 2 and Figure 4 show the AG News results. Mistral scores $F1 = 0.870$ with just $k = 12$ demonstrations. DistilBERT reaches $F1 = 0.826$ at $n = 400$ but does not surpass the baseline within our tested range. Extrapolating the fine-tuning curve, breakeven would likely land somewhere around $n \approx 500\text{--}600$.

Model	Config	Macro F1
Mistral (few-shot)	$k = 12$	0.870 ± 0.030
DistilBERT	$n = 20$	0.441 ± 0.080
DistilBERT	$n = 100$	0.742 ± 0.010
DistilBERT	$n = 400$	0.826 ± 0.041

Table 2: AG News results. Breakeven not reached at $n = 400$; estimated $n^* \approx 500\text{--}600$.

The gap between AG News and GoEmotions comes down to how much prior knowledge Mistral already has. Distinguishing World from Sports

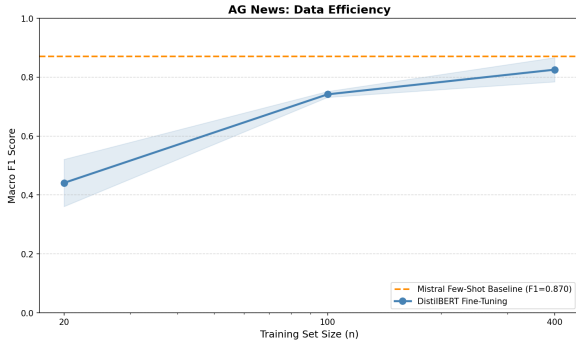


Figure 4: AG News Data Efficiency Curve. DistilBERT approaches but does not surpass the Mistral baseline within the tested range.

news is exactly the kind of broad topical categorization that is heavily represented in any large pre-training corpus, so Mistral gets it nearly right from just a few examples. Emotion recognition at 12-class granularity requires sensitivity to subtle linguistic cues that are harder to pick up from pre-training alone, which is why Mistral’s ceiling on GoEmotions is much lower and DistilBERT can catch up sooner.

4.3 SNLI: Natural Language Inference

Setup. SNLI (Bowman et al., 2015) requires classifying sentence pairs as entailment, neutral, or contradiction. We format inputs as “Premise: {premise} Hypothesis: {hypothesis}” and use 90 stratified test and validation examples.

Results. Table 3 and Figure 5 show that Mistral achieves $F1 = 0.785$ on SNLI, a strong baseline for this task. DistilBERT improves steadily from $F1 = 0.298$ at $n = 50$ to $F1 = 0.471$ at $n = 600$, but remains well below the baseline throughout. The standard deviations are notably large (0.10–0.12 at $n = 450$ and $n = 600$), which tells us that small training sets produce unstable NLI representations: some random seeds happen to land on a split that teaches the model reasonable inference patterns, others do not. In the extended sweep, DistilBERT reaches $F1 = 0.624$ at $n = 750$ and $F1 = 0.632$ at $n = 1200$, still short of Mistral’s 0.785. Breakeven on SNLI is probably somewhere beyond $n = 1500$.

NLI is structurally different from the other three tasks. Rather than classifying a single piece of text, the model must reason about the logical relationship between two sentences. With only 50 or 150 training examples, DistilBERT sees only about 17–50 examples of each relationship type, which is not enough to learn reliable pairwise rea-

Model	Config	Macro F1
Mistral (few-shot)	$k = 36$	0.785 ± 0.038
DistilBERT	$n = 50$	0.298 ± 0.057
DistilBERT	$n = 150$	0.286 ± 0.073
DistilBERT	$n = 450$	0.417 ± 0.119
DistilBERT	$n = 600$	0.471 ± 0.110

Table 3: SNLI results. Breakeven not reached; estimated $n^* > 1500$.

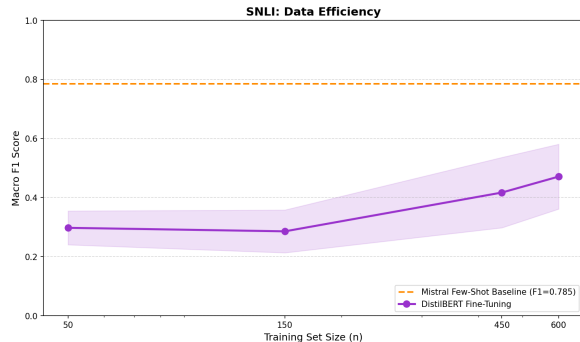


Figure 5: SNLI Data Efficiency Curve. DistilBERT remains well below the Mistral baseline even at $n = 600$, with wide confidence bands indicating high trial variance.

soning. Mistral, by contrast, has absorbed enormous amounts of logical and argumentative text during pre-training, giving it a natural advantage on inference tasks that does not erode easily.

4.4 CoLA: Linguistic Acceptability

Setup. CoLA (Warstadt et al., 2019) is a binary task from the GLUE benchmark (Wang et al., 2018): given an English sentence, determine whether it is grammatically acceptable. We split the training data 90/10 for train/validation and evaluate at $n \in \{50, 150, 450, 600, 750, 1000\}$.

Results. Table 4 and Figure 6 show the CoLA results. Mistral achieves $F1 = 0.720$. DistilBERT improves steadily up to $n = 750$ ($F1 = 0.636$), then flattens: $n = 1000$ gives $F1 = 0.632$, essentially the same as 750. The model appears to have hit a ceiling around $F1 = 0.63$ – 0.64 given its architecture and the data budget we tested.

The plateau is worth noting. On GoEmotions and SNLI, the fine-tuning curve is still climbing at the upper end of our data budget, suggesting that more data would eventually push DistilBERT past Mistral. On CoLA, the curve has leveled off, which suggests a different problem: grammatical acceptability judgments require sensitivity to syntactic patterns that may be genuinely difficult for

Model	Config	Macro F1
Mistral (few-shot)	$k = 36$	0.720 ± 0.056
DistilBERT	$n = 50$	0.420 ± 0.042
DistilBERT	$n = 150$	0.431 ± 0.068
DistilBERT	$n = 450$	0.606 ± 0.033
DistilBERT	$n = 600$	0.585 ± 0.033
DistilBERT	$n = 750$	0.636 ± 0.028
DistilBERT	$n = 1000$	0.632 ± 0.043

Table 4: CoLA results. Breakeven not reached within tested range; estimated $n^* > 1000$.

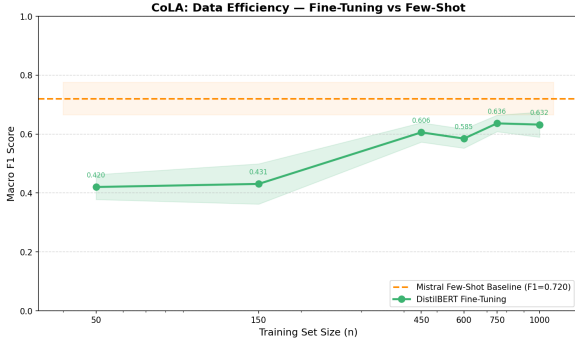


Figure 6: CoLA Data Efficiency Curve. DistilBERT improves steadily but plateaus around $n = 750$ and does not surpass the Mistral baseline within the tested range.

a 66M-parameter encoder to learn reliably from a few hundred examples, regardless of how much more data you add. Getting past Mistral on CoLA might require a stronger base model rather than just more labeled data.

4.5 Cross-Task Summary

Task	Classes	Mistral F1	Breakeven n^*
GoEmotions	12	0.370	≈ 450
AG News	4	0.870	> 400
SNLI	3	0.785	> 600
CoLA	2	0.720	> 1000

Table 5: Cross-task summary of Breakeven Points.

A few patterns stand out from looking at all four tasks together.

Label count is not the main driver. GoEmotions has the most classes (12) but the lowest breakeven point. CoLA has only two classes but the highest. What matters more is how well Mistral already handles the task out of the box.

LLM prior knowledge sets the bar. Mistral’s few-shot F1 ranges from 0.370 on emotion recognition to 0.870 on topic classification. Tasks where

pre-training data naturally covers the classification problem (news topics, logical inference, grammatical judgments) give Mistral a high starting point that DistilBERT has to work hard to exceed. Emotion recognition at fine-grained granularity is apparently not as well-covered by pre-training, which is why the gap is smallest there.

Fine-tuning efficiency depends on task structure. SNLI’s breakeven is pushed out not by a high Mistral baseline alone but also by the slow and noisy convergence of DistilBERT on a pairwise reasoning task. CoLA’s plateau suggests a model capacity limitation rather than a data limitation. These are qualitatively different reasons for a high breakeven point, and they suggest different solutions: more data helps for SNLI but may not help much for CoLA without changing the model.

5 Discussion

The most useful framing for these results is as a rough decision rule. If you have fewer than 150 labeled examples, few-shot prompting is the better choice across all four tasks we tested. Around 300 to 500 examples, fine-tuning starts to pay off if your task is a fine-grained classification problem where the LLM does not have strong priors. If your task involves something LLMs are already good at due to pre-training exposure, such as news topics, logical inference, or grammatical judgments, the fine-tuning investment pays off later and may require upwards of 1000 examples.

The sensitivity analysis on GoEmotions reinforces that the exact breakeven depends on how aggressively you prompt your LLM. At $k = 12$, DistilBERT can break even with around 300 training examples. At $k = 48$, you need closer to 450. Beyond $k = 48$, adding more prompt examples does not meaningfully raise Mistral’s score, so the bar stabilizes. In practice, this means you should figure out the best few-shot configuration you can realistically use before deciding whether annotation is worth it.

The emotion subset results add another wrinkle. Positive and negative emotion categories both require more data than the original mixed set, because swapping the label distribution changes the per-class sample density and may shift the relative difficulty of the distinctions the model needs to make. This is a caution against assuming that a breakeven measured on one label set will generalize cleanly to a related but different one.

Limitations. We used fast evaluation subsets of 90 to 120 examples across all tasks because of GPU time constraints on Google Colab. This introduces sampling noise that would not be present in a full-dataset evaluation. The high trial variance on SNLI in particular may partly reflect this: with only 90 test examples, a few classification errors can shift F1 by several points. The Mistral evaluations use greedy decoding with a fixed prompt template; different decoding strategies or prompt formats might shift the few-shot baseline somewhat. We also only tested DistilBERT-base-uncased and Mistral-7B-Instruct-v0.2, so we cannot say how these curves would look with a stronger encoder or a different LLM.

6 Conclusion

We measured the Breakeven Point between fine-tuned DistilBERT and few-shot Mistral-7B across four NLP tasks. The threshold is around 450 labeled examples for 12-class emotion recognition, somewhere above 400 for 4-class topic classification, above 600 for natural language inference, and above 1000 for binary grammatical acceptability. The task characteristics that push the breakeven higher are strong LLM prior knowledge, compositional reasoning requirements, and potential model capacity ceilings for the fine-tuned side.

For practitioners, the rough takeaway is: if you can label a few hundred examples and your task is one where large models do not already have strong pre-training coverage, fine-tuning a small model is likely worth it both for cost and performance. If your task is something well-covered by general pre-training, or you cannot afford more than a hundred labeled examples, prompting is probably the better call.

Future work should test this with parameter-efficient fine-tuning methods like LoRA, which could shift the breakeven point significantly by reducing the training signal needed to adapt the model. Testing additional LLMs of different sizes would also clarify how much of Mistral’s advantage comes from its specific instruction tuning versus raw model scale. Finally, building a cost-aware tool that takes task type, annotation budget, and expected inference volume as inputs and outputs a recommendation would be a natural practical extension of this work.

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